

Zamir Memon

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Summary

Grade 11 student with hands-on experience in full-stack web development, AI/ML, cloud computing, and security research. Published researcher in software security with a track record of self-directed learning across 19+ professional certification courses. Passionate about leveraging technology to solve real-world problems and build accessible, impactful solutions.

Technical Skills

- **Languages:** JavaScript, TypeScript, Python, Java, HTML, CSS
 - **Frameworks:** React, React Native, Expo, Node.js, Express, Django, Flask
 - **Databases:** MongoDB, Firebase/Firestore
 - **AI/ML:** Python for Data Science, AI Applications with Flask, Generative AI, Data Analysis
 - **Cloud & DevOps:** Docker, Kubernetes, OpenShift, Cloud Computing, Git/GitHub
 - **Security:** Bug bounty, cloud infrastructure analysis, vulnerability research, API testing
 - **Other:** Problem-solving, system design, REST APIs, microservices
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Education

High School Student (Grade 11) — India

Certifications

- **IBM Full Stack Software Developer** — 15 courses, Complete (*React, Node.js, Express, Django, Flask, Python, Docker, Kubernetes, Cloud Computing, AI Applications, Generative AI*)
 - **Introduction to Programming with Python & Java** — University of Pennsylvania, 4 courses, Complete (*Python Programming, Data Analysis, Java OOP, Data Structures*)
 - **JPMorgan Chase Software Engineering Simulation** — Forage (*Kafka, Spring Boot, REST APIs, database integration*)
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Research

Client-Side DRM Bypass Methods and a Proposed Server-Side Integrity Framework *Published Research Paper*

- Tested 3 real-world DRM bypass tools, all exploiting the same client-side verification flaw
- Analyzed bypass mechanisms and proposed a server-side integrity framework that respects modding and offline play

- Conducted security analysis using VirusTotal and reverse engineering techniques
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Projects

Bug Bounty Research — Indian EdTech Platform (Confidential)

- Conducted full security assessment of an Indian EdTech platform serving 4.2M+ users
- Identified critical vulnerabilities including unprotected CDN, exposed debug logs, API credential leaks, and IDOR flaws
- Mapped complete cloud infrastructure (AWS, CloudFront, S3, ALB) and documented attack surface

School Timetable Management System — Full Stack (MERN)

- Built a full-stack MERN web app (React 19, Node.js/Express 5, MongoDB/Mongoose) that automatically generates conflict-free school timetables
- Designed a constraint-based scheduling algorithm handling teacher quotas, max periods per day, no back-to-back conflicts, and fair distribution across multiple classes
- Implemented JWT-authenticated REST API with bcrypt password hashing, PDF export (PDFKit), Excel import/export, and drag-and-drop UI
- Deployed live on Railway with MongoDB Atlas

WalkWars — Territory-Conquest Fitness Game (React Native/Expo)

- Built a real-time, location-based game where walking in the real world claims map tiles on OpenStreetMap; submitted to Shipaton 2026 (RevenueCat global hackathon)
- Implemented a custom geo engine: geohash viewport queries, haversine distance, polygon area/simplification (Douglas–Peucker), and ray-casting point-in-polygon for claim/steal/overlap logic
- Engineered anti-cheat with GPS speed filtering (≤ 8 m/s) and mock-location rejection
- Integrated Firebase Auth + Firestore (users, territory tiles, leaderboard) with background location tracking via Expo TaskManager

JPMorgan Chase Software Engineering Simulation

- Built a Kafka-integrated Spring Boot microservice for real-time transaction processing
- Implemented transaction validation and persistence with Spring Data JPA and H2 database
- Connected to external REST Incentive API and developed balance query endpoints

Portfolio Website

- Built a Netflix-style portfolio using HTML, CSS, and JavaScript
 - Implemented interactive UI with popups, animations, and smooth scrolling
 - Showcases projects, skills, and certifications
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References available upon request